



ALLOYS — FULL MIND MAP

Topic-wise · Fully Explained · Hinglish · SSC / BPSC / State PCS

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1. Alloy Basics & Duralumin

- Alloy do ya zyada elements ka metallic mixture hota hai jise strength, hardness, corrosion resistance ya dusre useful properties improve karne ke liye banaya jata hai. Copper khud ek pure element/metal hai; Steel, Brass aur Bronze alloys hain. [Q1]
- Aircraft aur train-coach ke lightweight parts mein Aluminium-based alloys useful hote hain. Duralumin Aluminium-based alloy hai jo pure Aluminium se zyada strong hote hue bhi low-density rakhta hai. [Q2]

2. Amalgam — Mercury-based Alloys

- Amalgam ka basic rule: Mercury (Hg) + kisi dusri metal ka alloy/combination = amalgam. Isliye Mercury–Zinc amalgam hai. [Q3, Q5]
- Iron Mercury ke saath normal conditions mein readily amalgam nahi banata; isi standard exam fact ke karan Iron ko notable exception maana jata hai. [Q4]

3. Solder & Bronze

- Traditional soft solder mainly Tin (Sn) + Lead (Pb) alloy hota hai; low melting point ki wajah se metal pieces/components ko join karne mein use hota hai. [Q6]
- Bronze ka classic composition Copper + Tin hai. Source approximate 88% Cu + 12% Sn deta hai; exam mein “Bronze = Cu + Sn” core fact yaad rakho. [Q7, Q8]

⚠ EXAM TRAP — Solder/Bronze ke percentages fixed universal values nahi

- ▶ Solder ki exact Sn–Pb ratio application/grade ke hisab se vary kar sakti hai; source Q6 ek particular composition (~32% Sn, 68% Pb) deta hai.
- ▶ Bronze bhi ek alloy family hai; Cu–Sn percentage grade ke saath vary karta hai. Competitive exam ke liye composition pair sabse important hai.

4. German Silver — Name vs Composition

- German Silver (Nickel Silver) ka standard exam composition Copper + Zinc + Nickel hai. Naam mein “Silver” hone ke bawajood actual Silver 0% hota hai. [Q9, Q10, Q11]
- German Silver mein Aluminium constituent nahi hai. Matching recall: German Silver→Nickel, Solder→Tin, Bleaching Powder→Chlorine, Hypo (sodium thiosulphate)→Sodium. [Q9, Q12]

⚠ EXAM TRAP — German Silver mein Silver nahi hota

- ▶ German Silver ka naam appearance se aaya hai, composition se nahi: Cu + Zn + Ni.
- ▶ Q10/Q11 numerical options mein correct concept 0% Silver hai; agar 0% option na ho to “None of these” select hoga.

5. Brass, Bronze, Gunmetal & Copper-alloy Comparison

- Source comparison ke hisab se Bronze mein Copper percentage sabse zyada (~88%) diya gaya hai; Brass ~68–71% Cu, German Silver ~50–60% Cu aur Gunmetal ~85% Cu ke around bataya gaya hai. [Q13]
- Brass = Copper + Zinc; Bronze = Copper + Tin; German Silver = Copper + Zinc + Nickel. In teeno mein common base metal Copper hai. [Q14, Q15, Q16, Q17]
- Brass aur Bronze dono Copper alloys hain. Invar primarily Iron + Nickel alloy hai, jabki Nichrome mainly Nickel + Chromium (often Iron) alloy hota hai. [Q18]

⚠ EXAM TRAP — Alloy percentages grade ke saath change hote hain

- ▶ Q13 mein diye Cu percentages exam-source approximations hain; industrial Brass, Bronze aur Gunmetal ke multiple grades/compositions hote hain.
- ▶ Pair/composition questions mein exact percentage se zyada base-metal + major alloying element yaad rakhna safer hai.

6. Brass Tarnish & Statement-based Concepts

- Statement set mein Brass = Cu+Zn aur Amalgam = Hg-containing alloy correct hain. Magnetite Fe_3O_4 Iron ka ore hai, Aluminium ka nahi; traditional photography mein AgBr important light-sensitive compound hai, KNO_3 nahi. [Q19]
- Source ke according Brass ka dark/faded tarnish Hydrogen Sulphide (H_2S) se relate kiya gaya hai: Brass ka Copper sulphur species se react karke dark copper-sulphide type layer bana sakta hai. [Q20]

⚠ EXAM TRAP — Brass tarnish ko sirf ek gas tak oversimplify mat karo

- ▶ Real atmosphere mein oxygen, moisture, CO_2 aur sulphur compounds sab surface tarnishing/corrosion ko influence kar sakte hain.
- ▶ Lekin Q20 ke given options/source context mein H_2S intended answer hai.

7. Steel — Hardness, Carbon & Slag Uses

- Steel mein Carbon hardening agent ki tarah kaam karta hai; generally Carbon content badhne par hardness/strength badh sakti hai, saath hi ductility reduce ho sakti hai. [Q22]
- Steel slag iron/steel industry ka by-product hai. Iska use road-base/construction, acidic agricultural soils ke amendment aur cement/clinker raw-material applications mein kiya ja sakta hai. [Q23]

8. Stainless Steel — Chromium, Nickel & Composition

- Stainless steel ka key corrosion-resistant alloying element Chromium hai. Chromium surface par passive oxide film banane mein help karta hai, isliye rust/corrosion resistance improve hoti hai. [Q24, Q25, Q27, Q28]
- Nickel austenitic stainless-steel structure ko stabilize kar sakta hai, jisse common austenitic grades largely non-magnetic behaviour dikhate hain. Carbon hardness ko influence karta hai; Chromium corrosion resistance ka core element hai. [Q21]
- Exam composition recall: Stainless steel ka base Iron hota hai; Chromium essential hai, aur grade ke hisab se Nickel, Manganese, Carbon etc. bhi present ho sakte hain. Source Q29 Fe + Cr + Mn + C ko correct set deta hai. [Q29, Q30]
- Q26 ke source options mein Silicon ko “not used” key kiya gaya hai; Q31 BPSC-format mein strict two-metal options ko incomplete maan kar “more than one/none” key diya gaya hai. [Q26, Q31]

⚠ EXAM TRAP — Stainless Steel ek single fixed recipe nahi hai

- ▶ Har stainless steel non-magnetic nahi hota: austenitic grades generally non-magnetic, lekin ferritic/martensitic grades magnetic ho sakte hain.
- ▶ Silicon kuch stainless grades mein small amount mein ho sakta hai; Q26 ek simplified exam-source key hai.
- ▶ Q24 mein Chromium intended answer hai, lekin high-temperature/tool-steel properties mein W, Mo, V jaise alloying elements bhi important ho sakte hain.

9. Steel Carbon Range — Exam-safe Recall

- Source ke exam range ke hisab se ordinary steel mein Carbon approximately 0.1–2% by weight maana gaya hai; Carbon content steel ke hardness, strength aur ductility ko strongly affect karta hai. [Q32]

⚠ EXAM TRAP — Steel ka carbon range source-dependent hota hai

- ▶ Different standards/definitions steel ko roughly ~0.02% se ~2.1% C tak describe kar sakte hain; Q32 ke options mein 0.1–2% intended exam answer hai.
- ▶ Bahut high Carbon content par material cast-iron range ki taraf chala jata hai; exact boundary standard/composition par depend karti hai.

✓ Source-question mapping preserved: Q1–Q32